Satellite evidence for cooling assessments

HeatShield 813

UrbanCool AI, United Arab Emirates

Mussafah, Abu Dhabi: 1 km², 100 cells, one date

Official theme
Urban Expansion, Land Use Change & Heat Risk

The problem and who is affected

Heat varies across urban neighbourhoods

Exposed roofs, paved areas and limited vegetation create different surface conditions within the same city.

Affected communities

People who work outdoors or use exposed public spaces in hot urban districts.

Municipal decision gap

Limited cooling budgets need local evidence about where to inspect and what to assess.

This project measures surface temperature at satellite overpass times.

The business use case

End user

Municipal planning and environmental teams

Decision

Which locations should receive a site assessment for cooling measures?

Evidence supplied by the PoC

Observed surface heat
Broad vegetation signals
Dates and data provenance
Reason for each priority

Candidate measures: planting, shade, reflective roofs and cooler pavements.

Current-practice assumption: broad plans and separate imagery or site checks. Planner interviews are still needed.

Data used in the reproducible example

Input	Observation and role
Landsat 8, Collection 2 Level-2	11 June 2026 at 06:51 UTC Observed land surface temperature
Sentinel-2B, Level-2A	11 June 2026 at 06:46 UTC Six reflectance bands and spectral indices
Independent study window	100 cells at 100 m spacing 1 km × 1 km rectangle in Mussafah

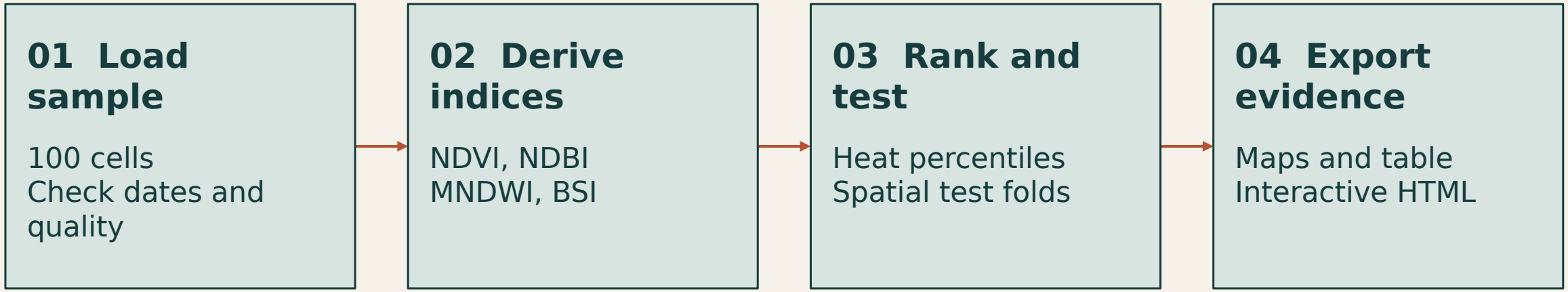
Hyperspectral status

No hyperspectral imagery contributes to these results. Satellite 813 access remains unconfirmed.

Sample starts after calibration and quality masking. Exact scene IDs and licences are in the README.

The implemented workflow

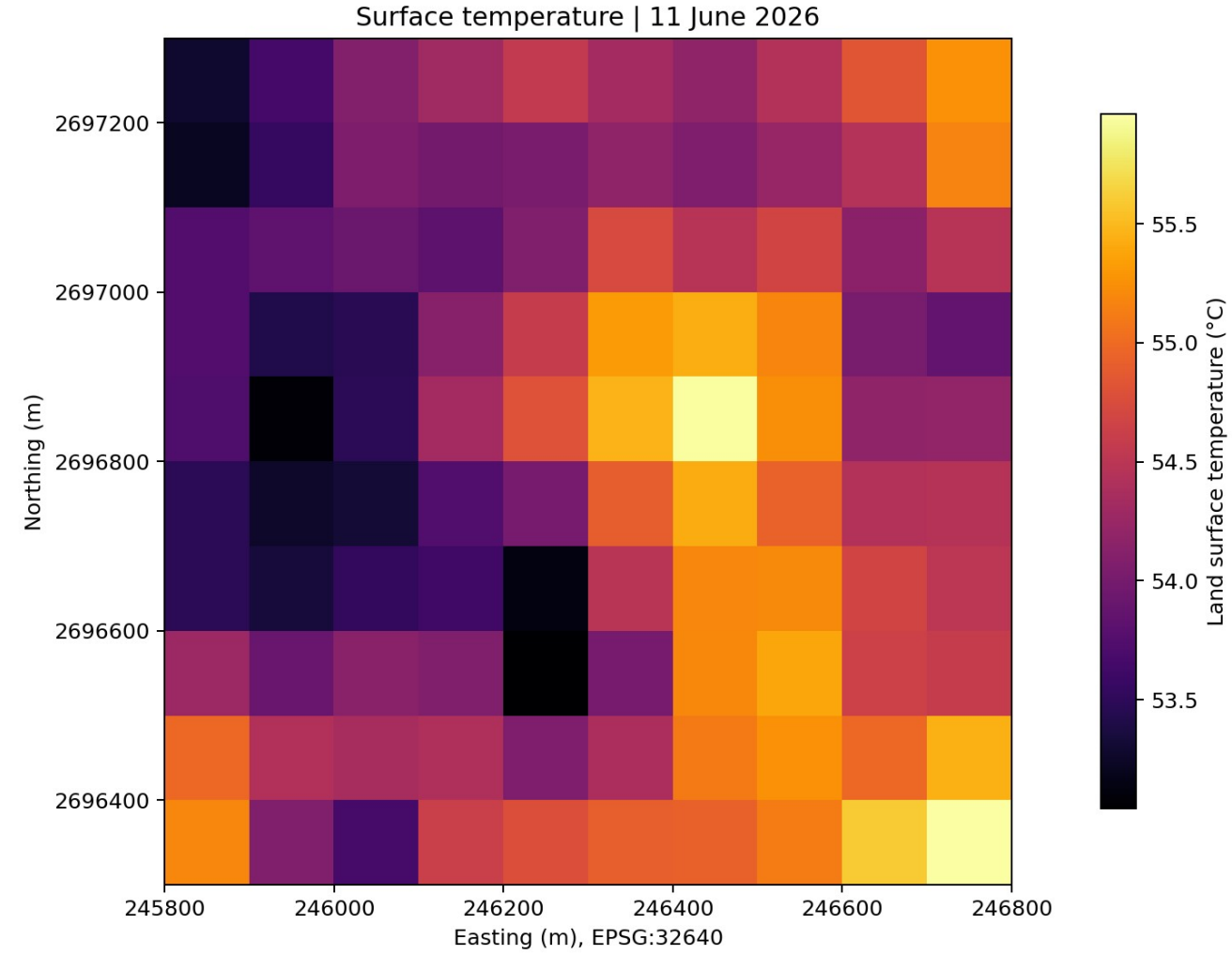
The notebook runs from a small, prepared satellite sample.



Mean baseline, linear regression and random forest share the same four spatial test folds. Seed = 42.

Raw-scene acquisition and preprocessing sit outside this offline example.

Observed surface temperature



11 June 2026

53.04–55.96 °C

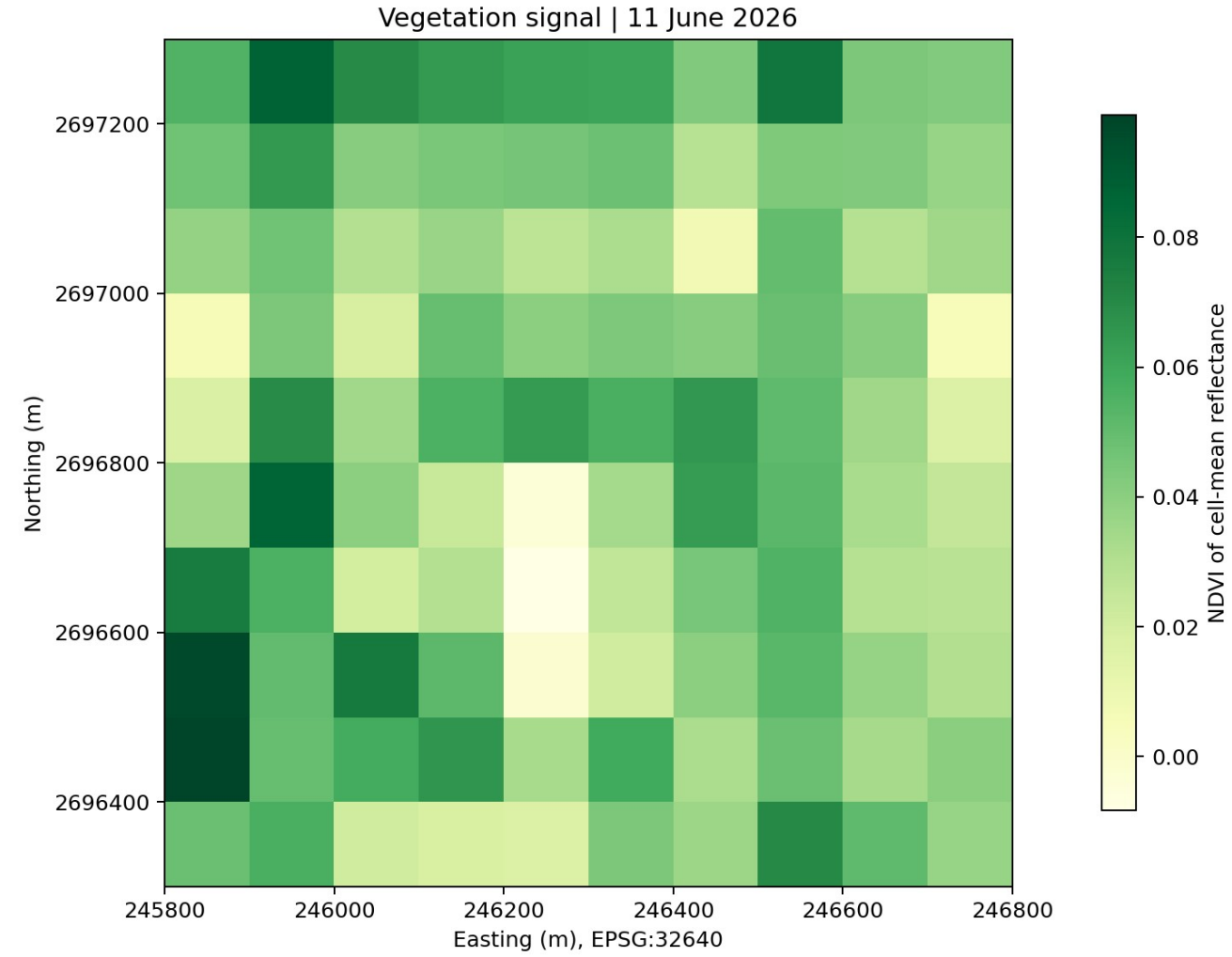
Observed range

54.36 °C

Mean across 100 cells

One satellite overpass.
Persistence is untested.

Vegetation signal across the sample



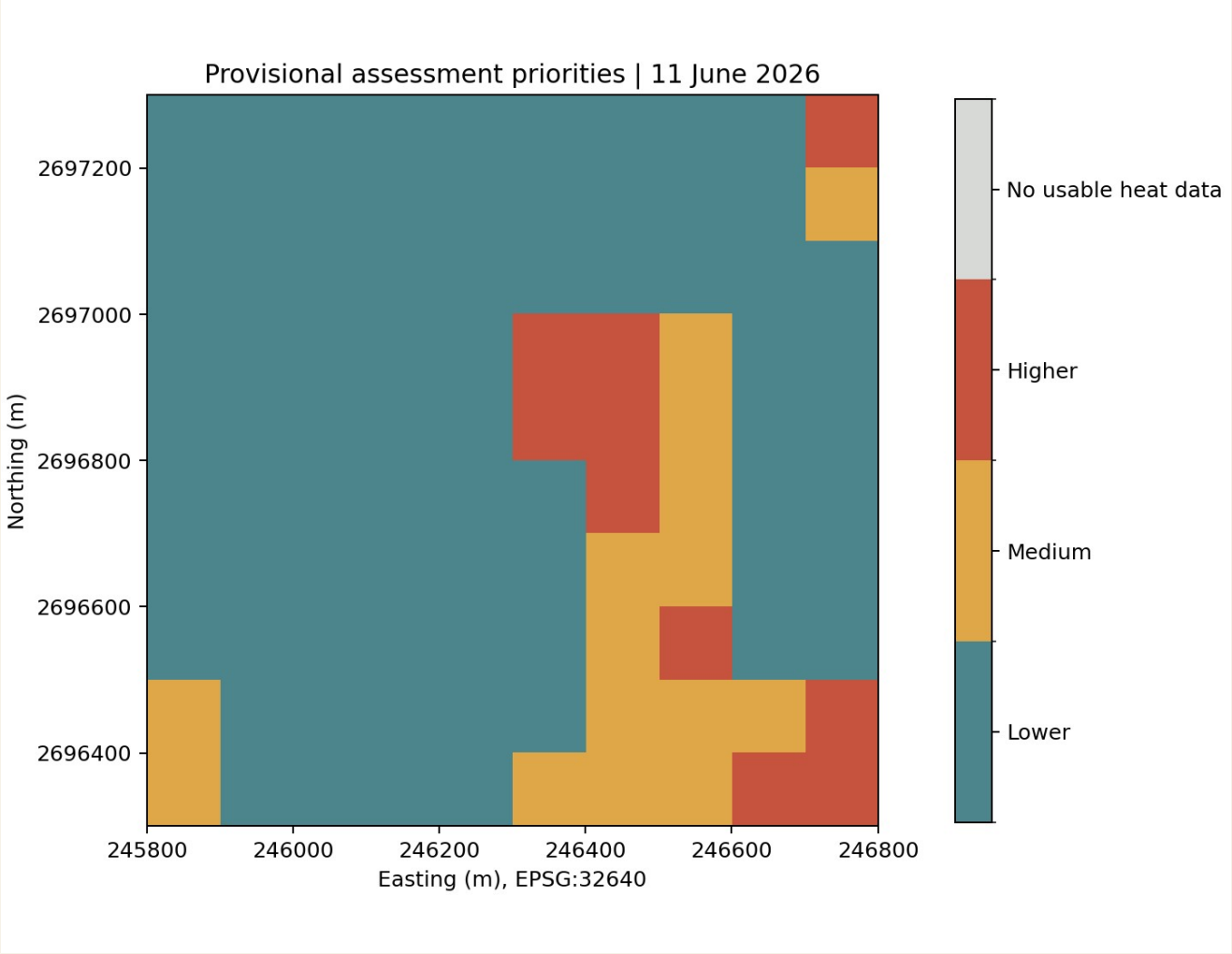
NDVI

A broad signal of vegetation condition

The analysis also derives NDBI, MNDWI and BSI from the same six bands.

Indices alone do not verify roof or pavement materials.

Provisional cooling-assessment priorities



Within-window heat ranking

10 higher
15 medium
75 lower

Higher: score ≥90
Medium: 75 to <90
Lower: score <75

Thresholds are exploratory.

Evidence behind a mapped priority

window_r09_c09

Evidence field	Actual output
Surface temperature	55.96 °C
Relative to window median	+1.64 °C
Heat score and priority	99.5 / 100, Higher
Observation date	11 June 2026

Candidate assessments

Shade for outdoor use

Planting feasibility and water needs

Reflective roofs, after verifying roofs

Cooler pavements, after verifying paving

Source: results/priority_table.csv. Each candidate action needs a site check.

Interactive HTML supports cell selection, layer changes and CSV export.

Spatial validation shows limited generalisation

Model	Mean MAE	Mean RMSE
Mean baseline	0.817 °C	0.934 °C
Linear regression	0.966 °C	1.183 °C
Random forest	0.746 °C	0.878 °C

Validation design

Four spatial folds
Training cells >200 m from test-cell centres

Random forest improves mean MAE by 0.071 °C. Linear regression is worse.

All models have negative mean fold R^2. More dates and independent sites are needed.

Unweighted fold averages. One observation date prevents temporal validation.

Scope and limitations

One date	No hotspot persistence or held-out-date test.
Independent window	The 1 km² demonstration does not represent all of M11–M12.
Prepared satellite values	Raw-scene calibration and masking are documented but not rerun here.
Surface associations	No direct human heat exposure, material validation or cooling-effect estimate.

Exact-sector expansion needs an authorised boundary and independent ground evidence.

Expected impact

A clearer basis for local cooling assessments

Planners could use traceable surface evidence to select locations for field checks and compare candidate measures.

Longer-term value

A repeatable workflow for comparable heat-adaptation decisions across Arab cities.

Next steps for incubation

01	**Secure the pilot scope**	Obtain an authorised M11–M12 boundary and confirm user needs.
02	**Expand the evidence**	Add observation dates, test persistence and reserve dates for validation.
03	**Test on the ground**	Check surface labels, candidate actions and operational usefulness.
04	**Evaluate hyperspectral value**	Use suitable imagery only if access and additional benefit are demonstrated.

Training and mentorship will help the team test responsibly and continue beyond the competition.

UrbanCool AI team

All five members developed the project collaboratively.

Member	Role
Nyar Adil Babiker Abdelrahman	Team lead and co-developer
Shahd Ehab Abdulmalek Eshaq	Co-developer
Nour Alhoda Abdulrahman Alahmed	Co-developer
Yosra Ghneem	Co-developer
Ruba Babikir Ali Mohamed	Co-developer

United Arab Emirates